

Xiaohui Wang

TODO: personal webpage
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EDUCATION

Carnegie Mellon University

Summer Intern at Carnegie Mellon University Database Group, advised by Professor Andy Pavlo

Pittsburgh, USA

June. 2025 - Present

Zhejiang University

Bachelor of Engineering in Computer Science (expected in 07.2026)

Hangzhou, China

Sept. 2022 - Present

GPA (overall): **90.40/100**; Ranking: **10%**

Core courses: Compiler Design(A), Software Engineering(A), Computer Networks(A), Computer Architecture(A), Object-Oriented Programming (A), Mathematical Statistics(A+), Natural Language Processing(A+), Discrete Mathematics(A+)

First Prize in Zhejiang University(3%)

Oct. 2023

Third Prize in Zhejiang University(15%)

Oct. 2024

RESEARCH EXPERIENCES

Summer Research Intern

June. 2025 - Present

Carnegie Mellon University Database Group, advised by Professor Andrew Pavlo

<https://db.cs.cmu.edu/>

- Developed an autonomous proxy-based LLM query tuning framework that enables non-invasive database optimization by intercepting and adapting queries through a lightweight proxy layer, allowing performance improvement without modifying the underlying DBMS
- Demonstrated substantial performance improvements, leveraging historical query hints as prior knowledge and achieving 38.77% latency reduction on average over the baseline on the TPC-H benchmark, highlighting the potential of LLM-driven optimization in practical database workloads
- Delivered a fully functional prototype and a comprehensive experimental study on the TPC-H benchmark, providing insights into the functionality of the proxy and the LLM's potential for the query optimization; a paper based on this work is currently in preparation for submission to a VLDB Workshop

Research Intern

Nov. 2024 - Jan. 2025

Data Intelligence and Database Laboratory, Zhejiang University, advised by Professor Chen, Gang

<https://dilab-zju.github.io/>

- Developed an end-to-end data preprocessing and entity matching system, integrating data normalization, transformation, and BERT-based model fine-tuning to handle heterogeneous table schemas
- Achieved significant accuracy gains, improving entity matching precision by 4.5% (Average) and reducing preprocessing time through automated schema alignment
- Provided insights into model overfitting on small datasets and scalability bottlenecks on large ones, forming the basis for a forthcoming publication in data integration research

PROJECT EXPERIENCES

Product Comparison Website(with Python, Javascript, SQLite3)

Nov.2024-Dec.2024

B/S architecture design Course design at Zhejiang University

- Developed a product comparison platform using Python, JavaScript, and SQLite3, implementing a B/S architecture to manage users, products, and price history
- Enabled real-time price tracking and notifications, automatically fetching product prices from major e-commerce platforms via Selenium, and sending alerts through email and app push notifications
- Delivered a functional prototype with a persistent SQLite3 database, real-time scraping module, and automated reminder system, providing a practical solution for monitoring price fluctuations

Compiler Design Project(with C++)

Mar.2025-June2025

Compiler Design Course at Zhejiang University

- Designed and implemented a compiler project, independently developing two core modules: the Lexical & Syntax Analyzer and the Intermediate Code Generator

- Ensured accurate compilation, implementing lexical tokenization, syntax parsing, and intermediate representation generation, handling errors gracefully and supporting basic optimization rules
- Contributed high-quality test cases to the course assignments, enhancing the testing coverage and robustness of the compiler system

Industry Experience

Text to Videos Generation Models(with Pytorch, DiffSynth-Studio)

Hangzhou,China

Team member, Kolors-LoRA Tournament – 2nd Prize, First Place

Sept.2024-Sept.2024

- Developed a text-to-video generation workflow integrating ExVideo and Stable Video Diffusion to synthesize high-quality videos from textual prompts
- Enhanced model adaptability by fine-tuning the diffusion backbone with Kolors-style LoRA modules, achieving significantly improved stylistic consistency and visual fidelity
- Produced a curated dataset of diverse character images and generated stylized video outputs, earning 2nd Prize, First Place in the Kolors-LoRA Tournament

SKILLS AND OTHERS

Computer: Rust, Python, C, C++, Visual Studio Code, MySQL, Postgresql, Overleaf

Language: English (Fluent), TOFEL(105)

Interests: Basketball, Table Tennis, Hiking, Host